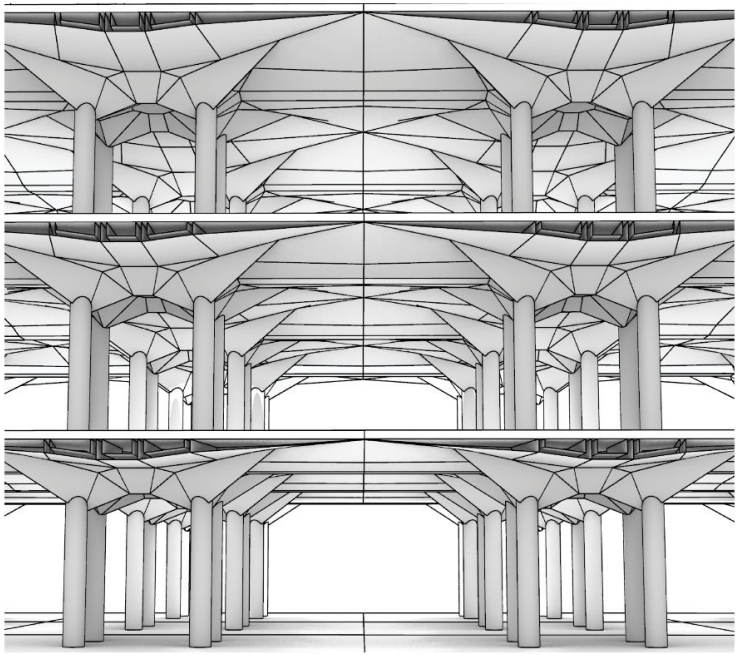


LITHIC PULSE

MArch Architectural Design I BPro I RCX 2025 - 2026

Bartlett School of Architecture I UCL

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Yashaswini Patel	25113495
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Pranav Lokhande	25142933

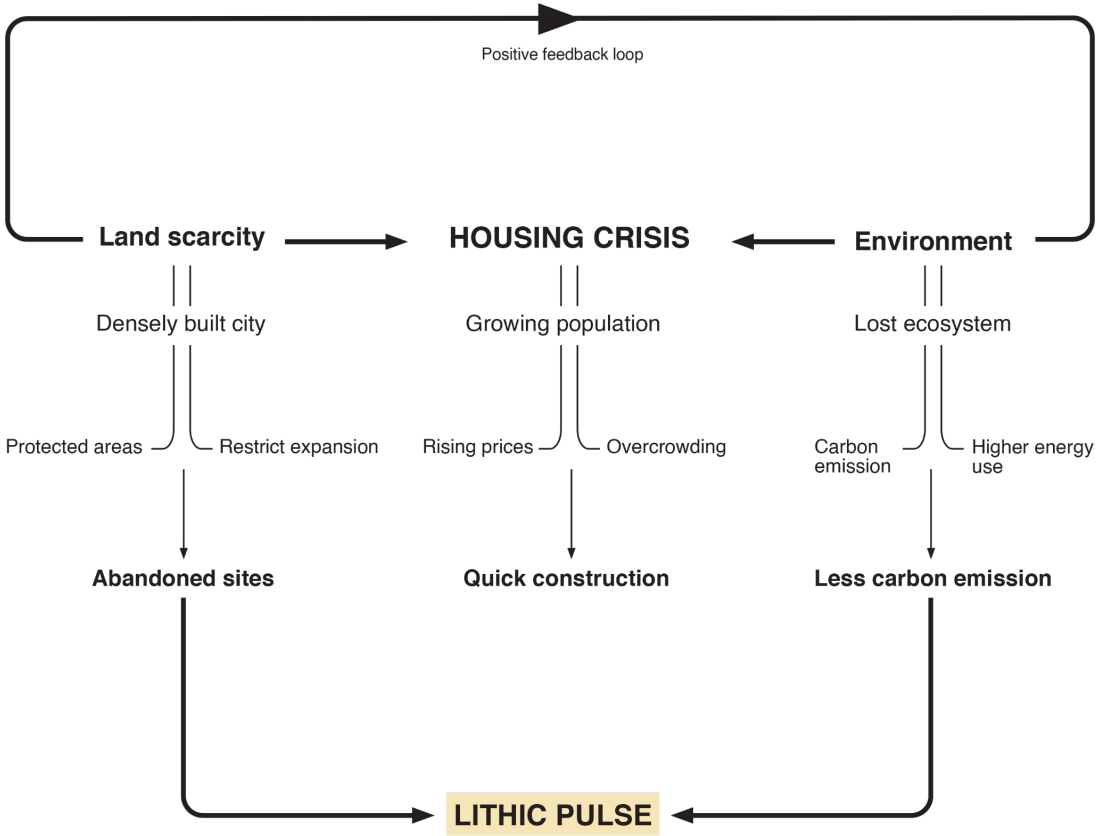


Figure 1. System diagram of London's housing crisis
Source: Lithic Pulse UCL Bartlett RCX

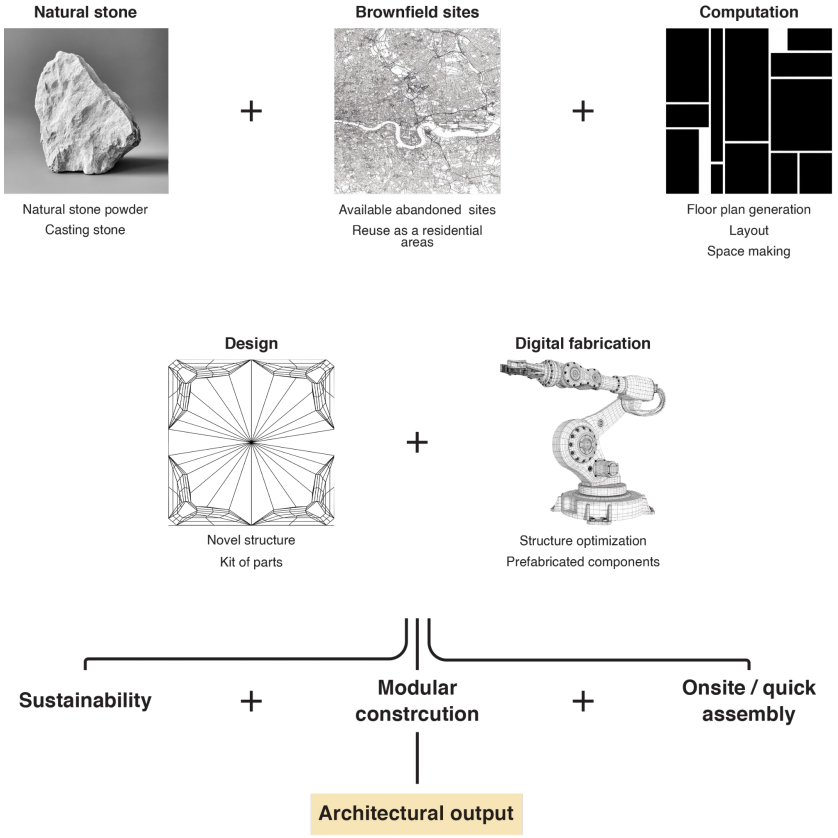


Figure 2. Solution framework for London's housing crisis
Source: Lithic Pulse UCL Bartlett RCX

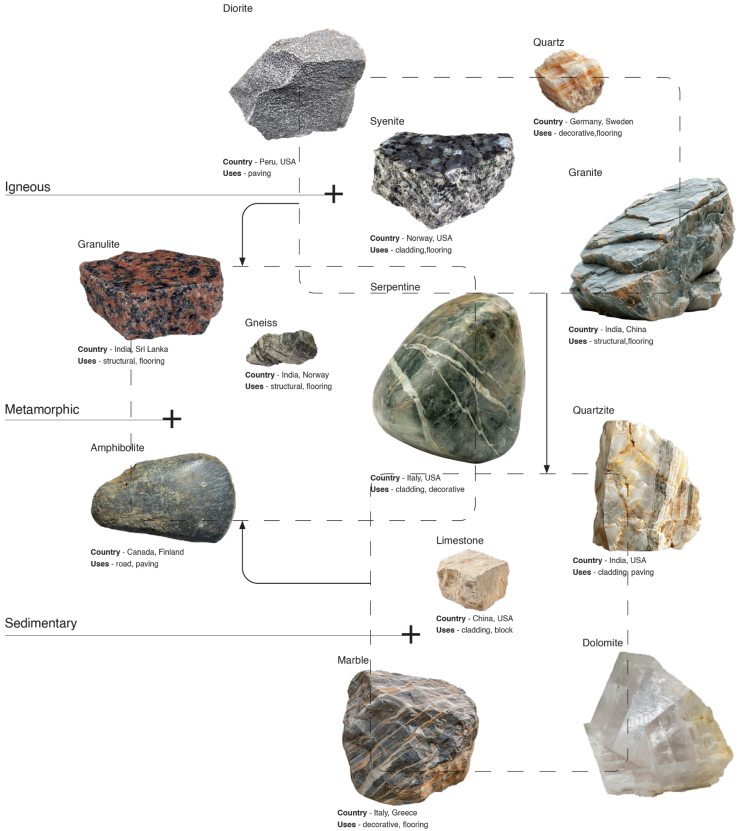
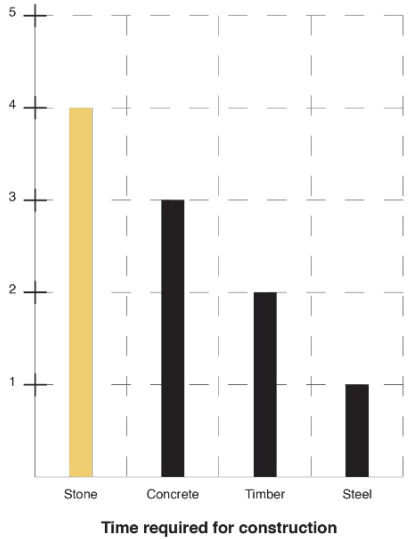
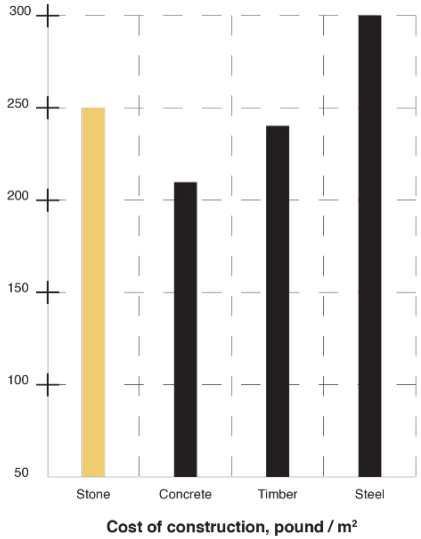
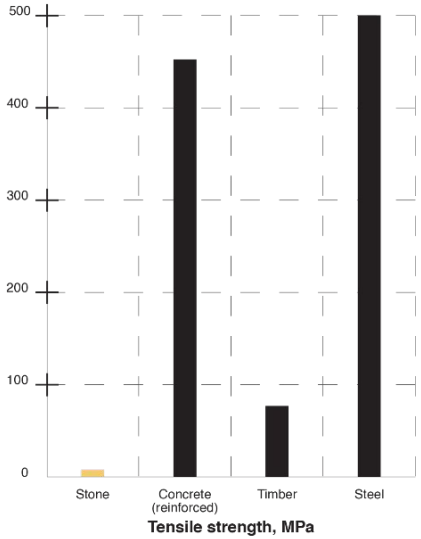


Figure 3. Natural stone
Source: Lithic Pulse UCL Bartlett RCX







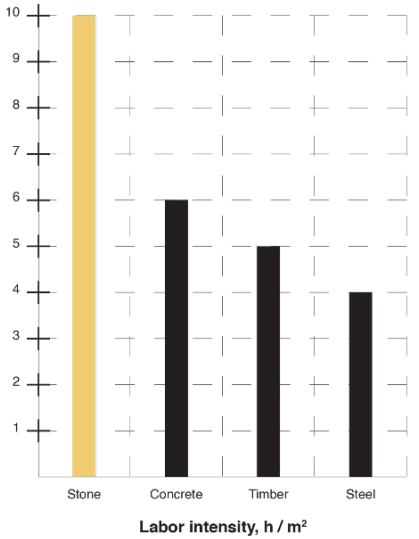


Figure 4. Why stone housing declined?
Source: Lithic Pulse UCL Bartlett RCX

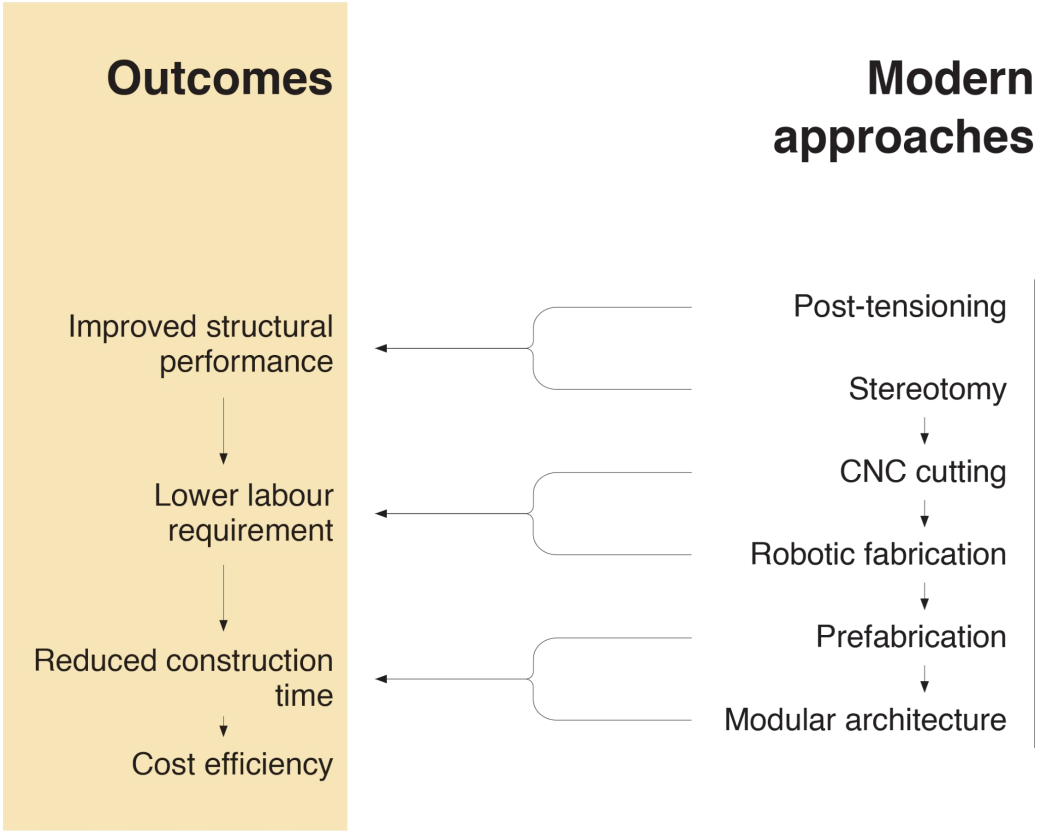


Figure 5. Improvements in stone construction
Source: Lithic Pulse UCL Bartlett RCX



Figure 6. Why cast stone?
Source: Lithic Pulse UCL Bartlett RCX

MATERIAL

The Material Paradox

Subtractive Stone versus Formative Concrete

Natural Stone

Beautiful and low in carbon but entirely subtractive. You must carve it to shape it which limits geometric freedom and creates massive waste.

Traditional Cement

Concrete allows us to cast complex shapes but heavily pollutes the environment. It offers ultimate geometric freedom but carries a devastating ecological cost.

A Formative Matrix

We engineered a pourable stone mixture. We wanted a material that pours freely into complex molds like concrete but retains the structural integrity and low carbon footprint of raw minerals.



First Casts

Limestone Powder and GGBS

The first 4 casts used limestone powder mixed with GGBS, plasticiser, basalt fibre, and water.

Different ratios were tested to find the right balance between binder and filler.

Simultaneously tested mould types.

Red sand moulds: Geometry lost.

Beach sand moulds: Imprecise.

PLA moulds: Clean geometry. Precision.

RESULT - BRITTLE

Marble dust 150g
GGBS 50g
Cement 50g
Basalt 1g
Superplasticizer 3ml



Marble dust 100g
GGBS 100g
Cement 5g
Basalt 0
Superplasticizer 2ml





Marble dust 200g
GGBS 20g
Cement 25g
Basalt 1g
Superplasticizer 0ml



Marble dust 50g
GGBS 150g
Cement 50g
Basalt 0
Superplasticizer 2 ml





Pivot

Cement Added.
Marble Replaces Limestone.

After the brittle failure, two definitive changes were made.

Added Portland cement to force fast, strong curing.

Marble possesses a higher silica content for better alkali activation, yielding a vastly stronger result and a whiter mineral finish.

More Basalt Fibers



Balanced -
Marble and GGBS



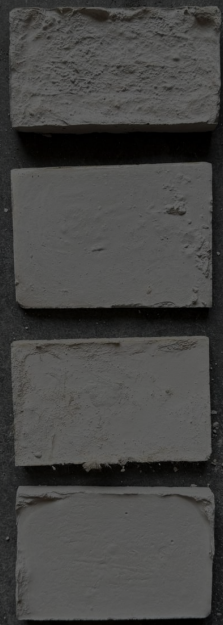
Balanced
Marble and GGBS



More superplasticizer



THE BREAKTHROUGH



Everyone assumed cast stone would be soft and rounded. It held every edge perfectly.

Iterative Load Testing

Finding the Chemical Balance

We tested our casts by stacking heavy PLA spools and water bottles directly on top of them. These physical loads revealed strict chemical limits. Mixes with excess basalt fibre became too rigid and snapped. Mixes flooded with plasticizer lost their binding strength and crumbled. The material only survived the weight when the ingredients were perfectly balanced.













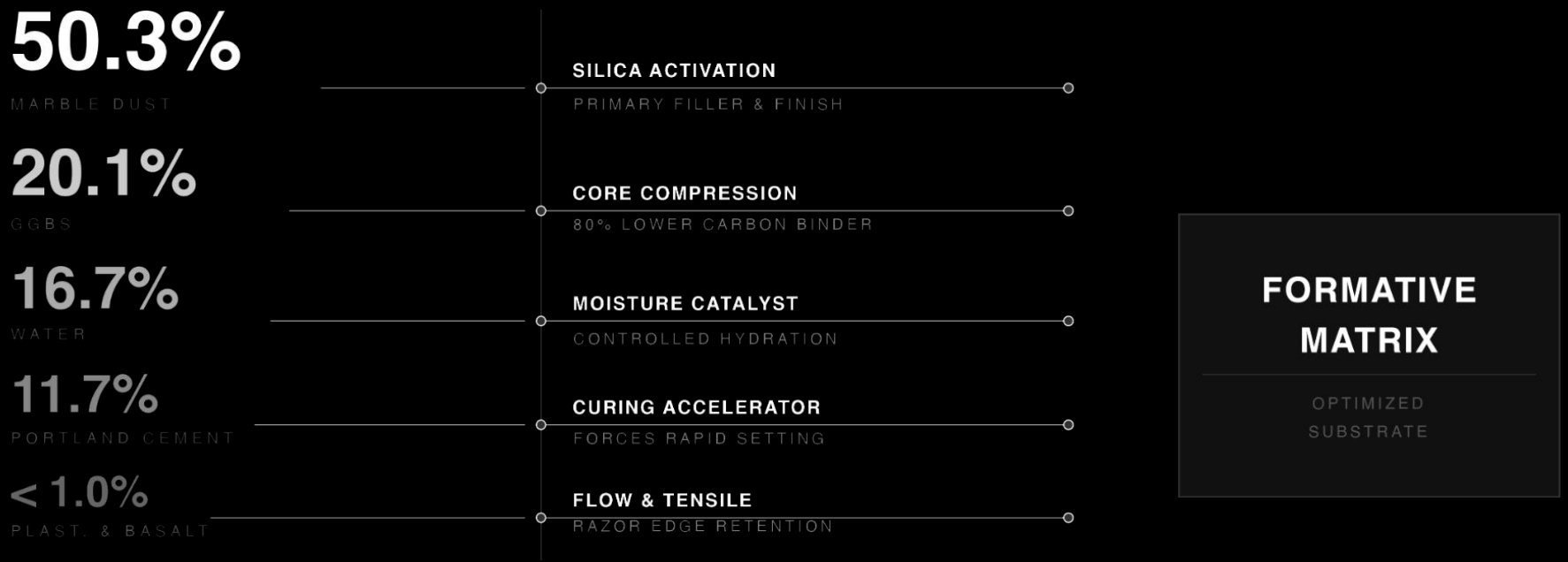


Figure 7. Parametric formulation - the receipt that doesn't break
Source: Lithic Pulse UCL Bartlett RCX

Pigmentation Dynamics

We are experimenting with material pigmentation within our community housing project to organically guide behavior through the architecture itself, rather than relying on signs or arrows. The challenge lies in striking a balance: large, uniform surfaces in shared public areas can feel cold and disorienting, while chaotic, overly saturated tones can induce stress and anxiety for residents.

To solve this, we treat material pigmentation like a visual volume dial. We turn the color and pattern intensity "up" in high-traffic public spaces where we want people to feel alert, energized, and moving. Conversely, we turn it "down" to muted, calming tones in semi-open, shared pockets to naturally signal residents to slow down, sit, converse, and feel safe.



STRUCTURAL DESIGN

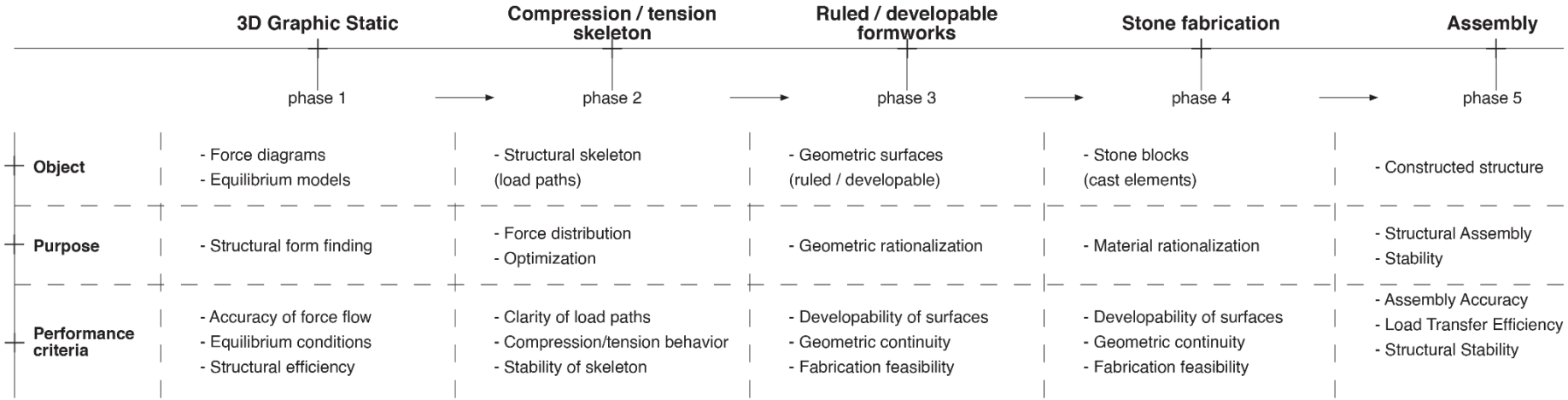
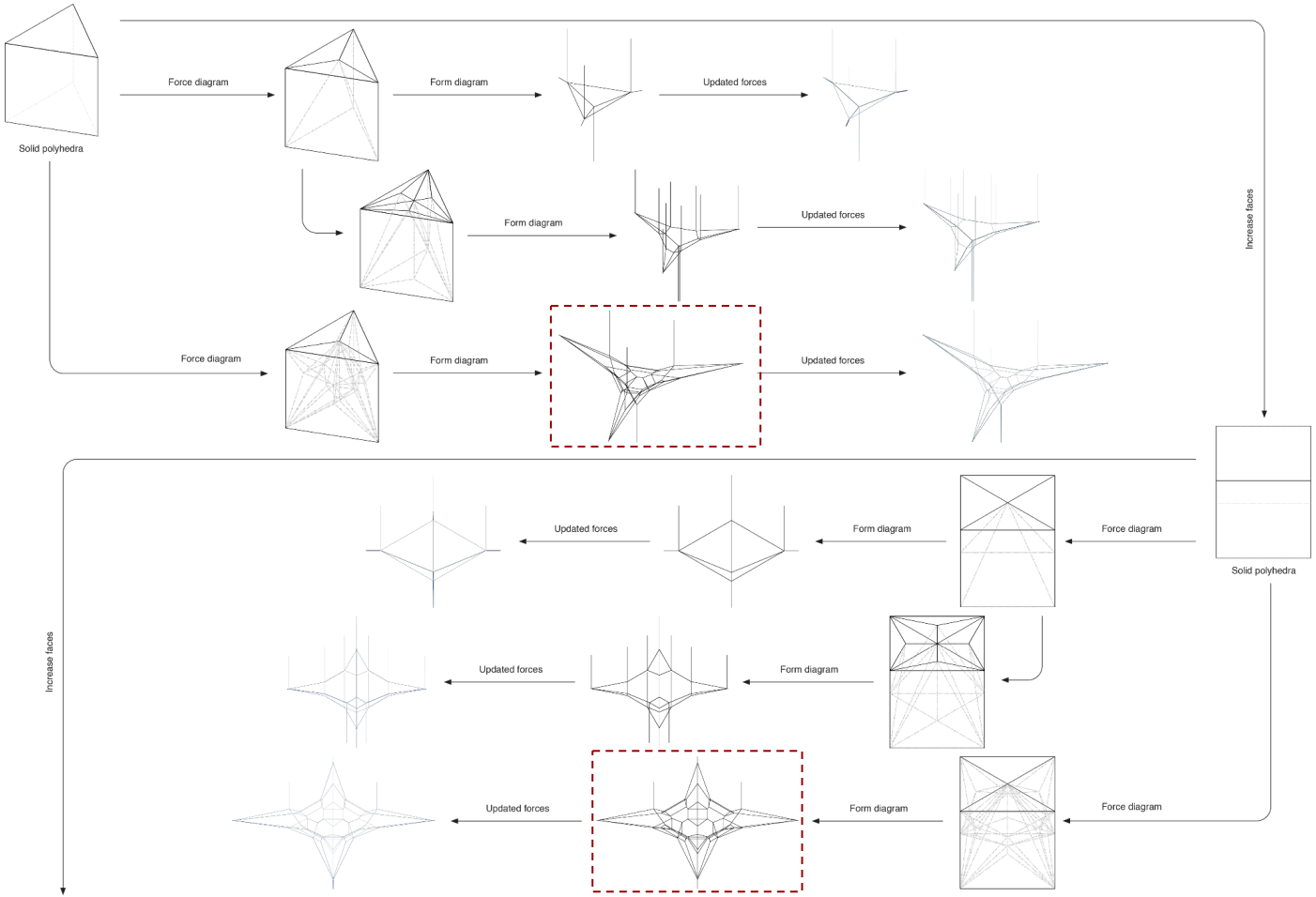


Figure 8. Design-to-assembly workflow
Source: Lithic Pulse UCL Bartlett RCX


Solid polyhedra
Force diagram
Form diagram
Updated forces
Force diagram
Form diagram
Updated forces
Force diagram
Form diagram
Updated forces
Increase faces
Solid polyhedra
Force diagram
Form diagram
Updated forces
Force diagram
Form diagram
Updated forces
Force diagram
Form diagram
Updated forces
Increase faces

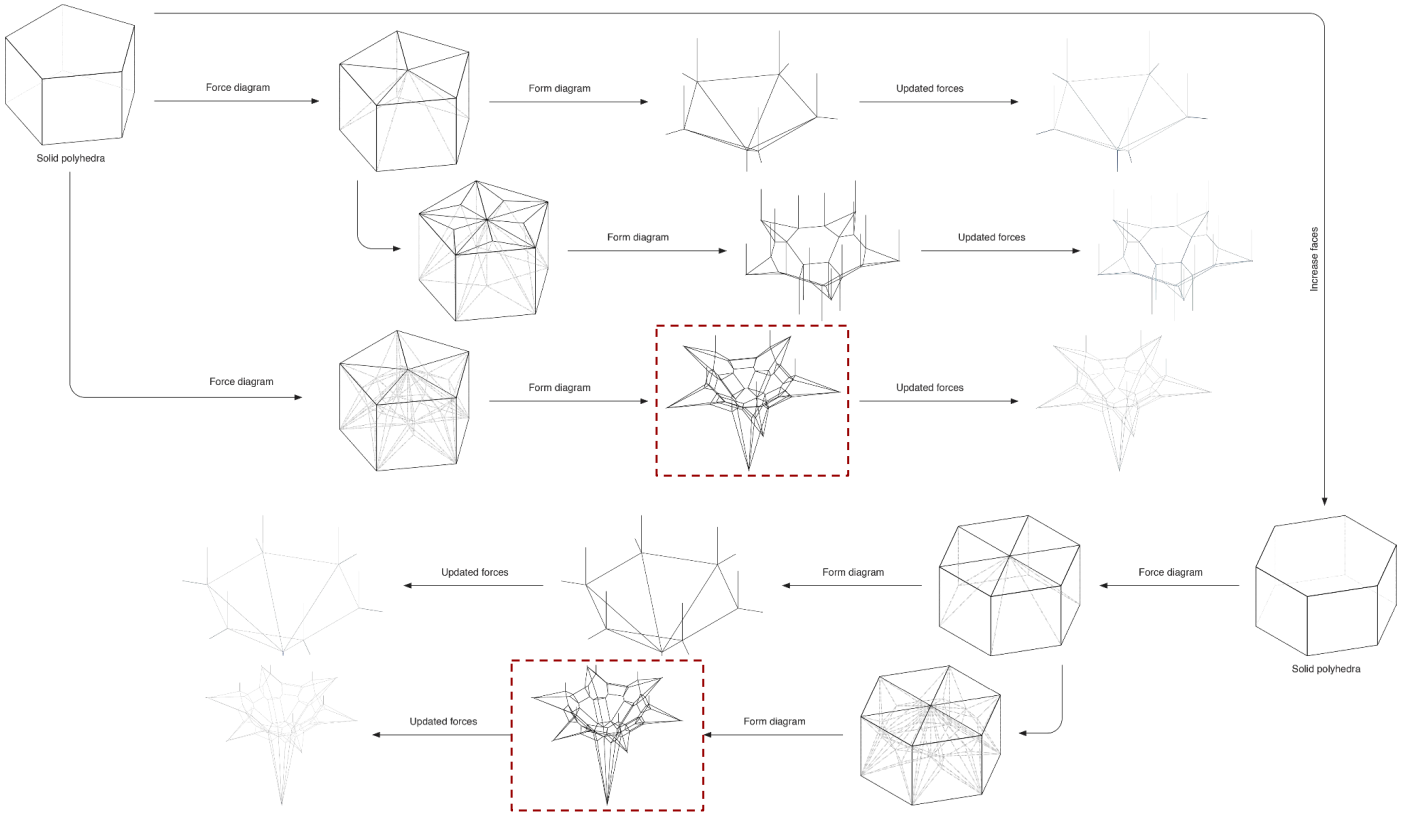
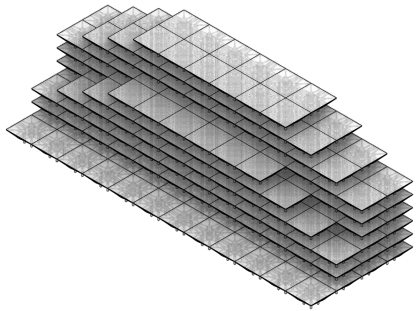


Figure 9. Structural form finding
Source: Lithic Pulse UCL Bartlett RCX

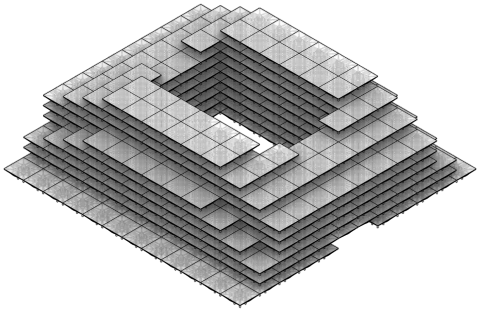
SPATIAL AGGREGATION FRAMEWORK

How
can spatial aggregation frameworks offer more socially resilient alternatives to standardized contemporary housing development?



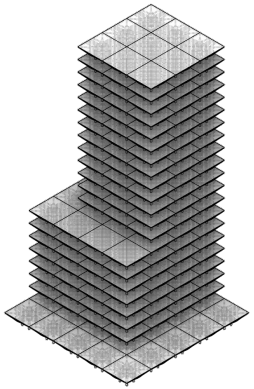
Linear planning

Advantages:
1. Offers a simple and efficient structural solution;
2. Facilitates clear apartment orientation;
3. Provides efficient circulation and communication systems;
Disadvantages:
1. Creates a monotonous urban silhouette;
2. Weakens the quality of open spaces;
3. Limits opportunities for social interaction.



Perimeter block planning

Advantages:
1. Strengthens the urban street edge;
2. Creates semi-enclosed internal courtyards;
3. Provides acoustic protection against external noise
Disadvantages:
1. Solar access within inner courtyards may be limited;
2. Lower floors can experience insufficient natural lighting;
3. Natural ventilation may be restricted.



Tower-type planning

Advantages:
1. Enables high-density development;
2. Provides panoramic views and visual permeability;
3. Optimizes land use efficiency;
Disadvantages:
1. Weakens the perception of human scale;
2. Increases overshadowing effects;
3. Reduces social interaction within communal spaces.

Figure 10. The primary residential planning typologies
Source: Lithic Pulse RCX

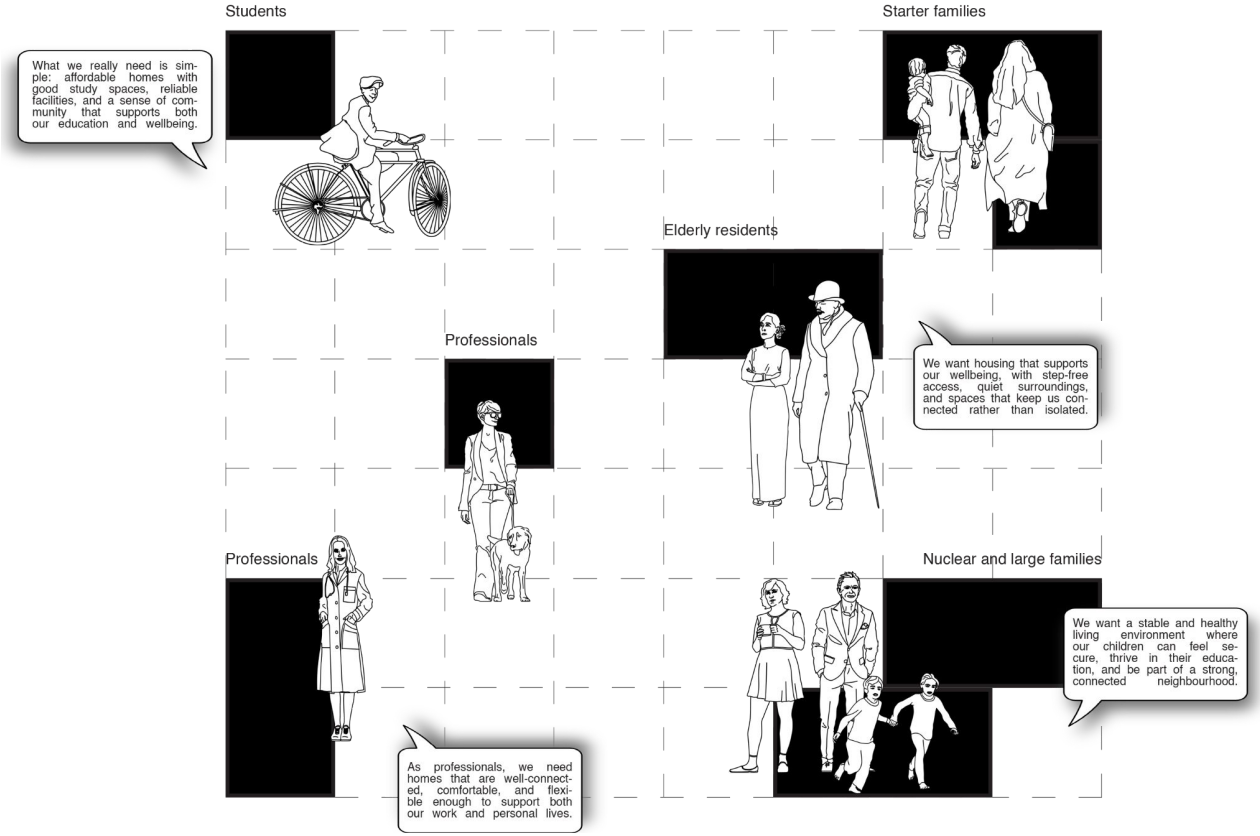


Figure 11. User profile diagram
Source: Lithic Pulse UCL Bartlett RCX

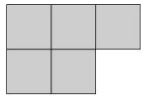
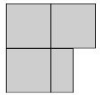
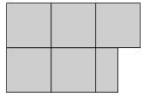
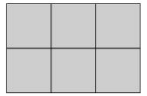
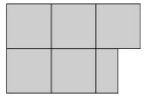
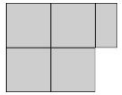
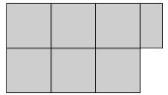
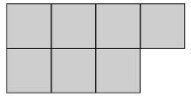
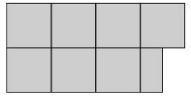
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1b2p	50 - **56***		1.5	3b5p	86 - **88***		2.5
				3b6p	95 - **96***		2.5
2b3p	61 - **64***		2.0	4b5p	90 - **88***		3.0
2b4p	70 - **72***		2.0	4b6p	99 - **104***		3.0
				4b7p	108 - **112***		3.0
				4b8p	117 - **120***		3.0

Figure 12. Design codes for the space planning (Left: UK Nationally Described Space Standards (NDSS); Right: Proposed spatial standards defined within the project)
Source: Lithic Pulse RCX

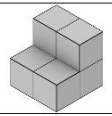
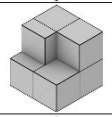
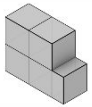
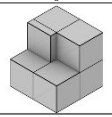
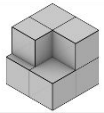
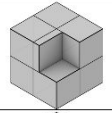
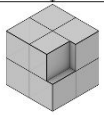
№ bedroom & bed spaces	2 storey unit (m2)	2 storey unit (grid)	Storage	№ bedroom & bed spaces	2 storey unit (m2)	2 storey unit (grid)	Storage
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1b2p	58 - **60***		1.5	3b5p	93 - **96***		2.5
				3b6p	102 - **104***		2.5
2b3p	70 - **72***		2.0	4b5p	97 - **100***		3.0
2b4p	79 - **80***		2.0	4b6p	106 - **108***		3.0
				4b7p	115 - **116***		3.0
				4b8p	124 - **124***		3.0

Figure 13. Design codes for the space planning (Left: UK Nationally Described Space Standards (NDSS); Right: Proposed spatial standards defined within the project)
Source: Lithic Pulse RCX

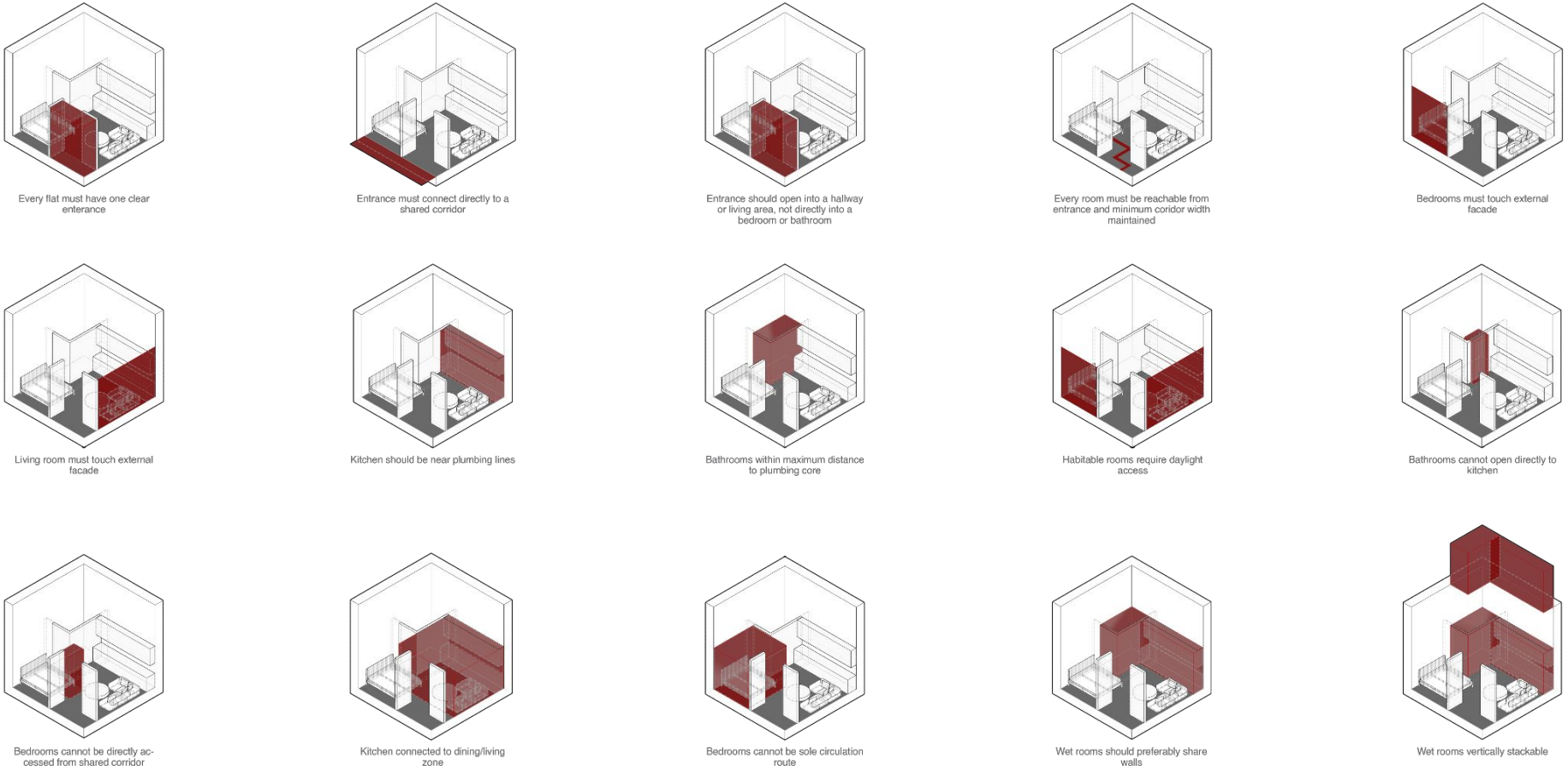


Figure 14. Design codes for the space planning
Source: Lithic Pulse RCX



Figure 15. Different grid aggregation and manual floor plans
Source: Lithic Pulse RCX

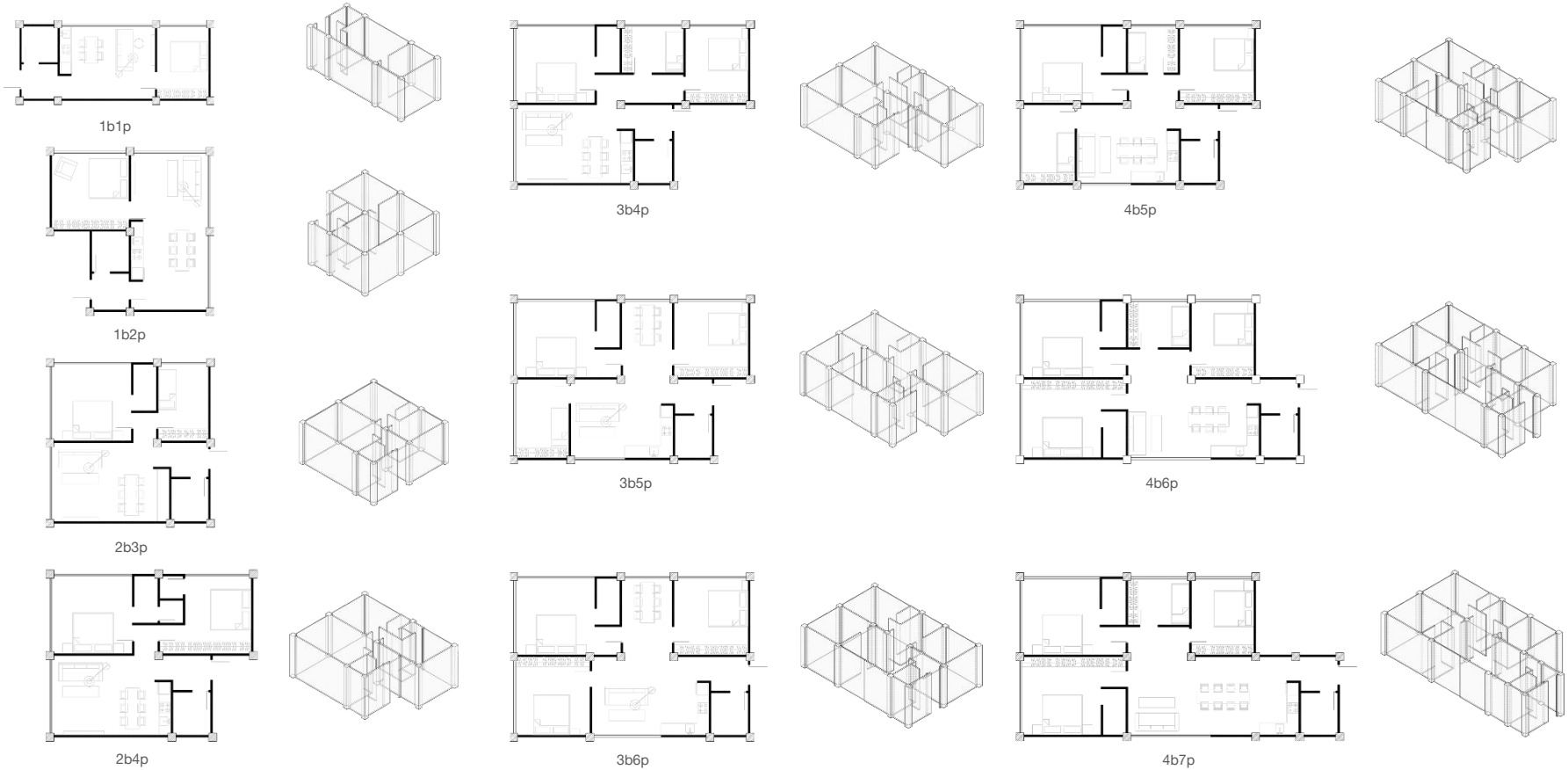


Figure 16. Spatially organized floor plans
Source: Lithic Pulse RCX

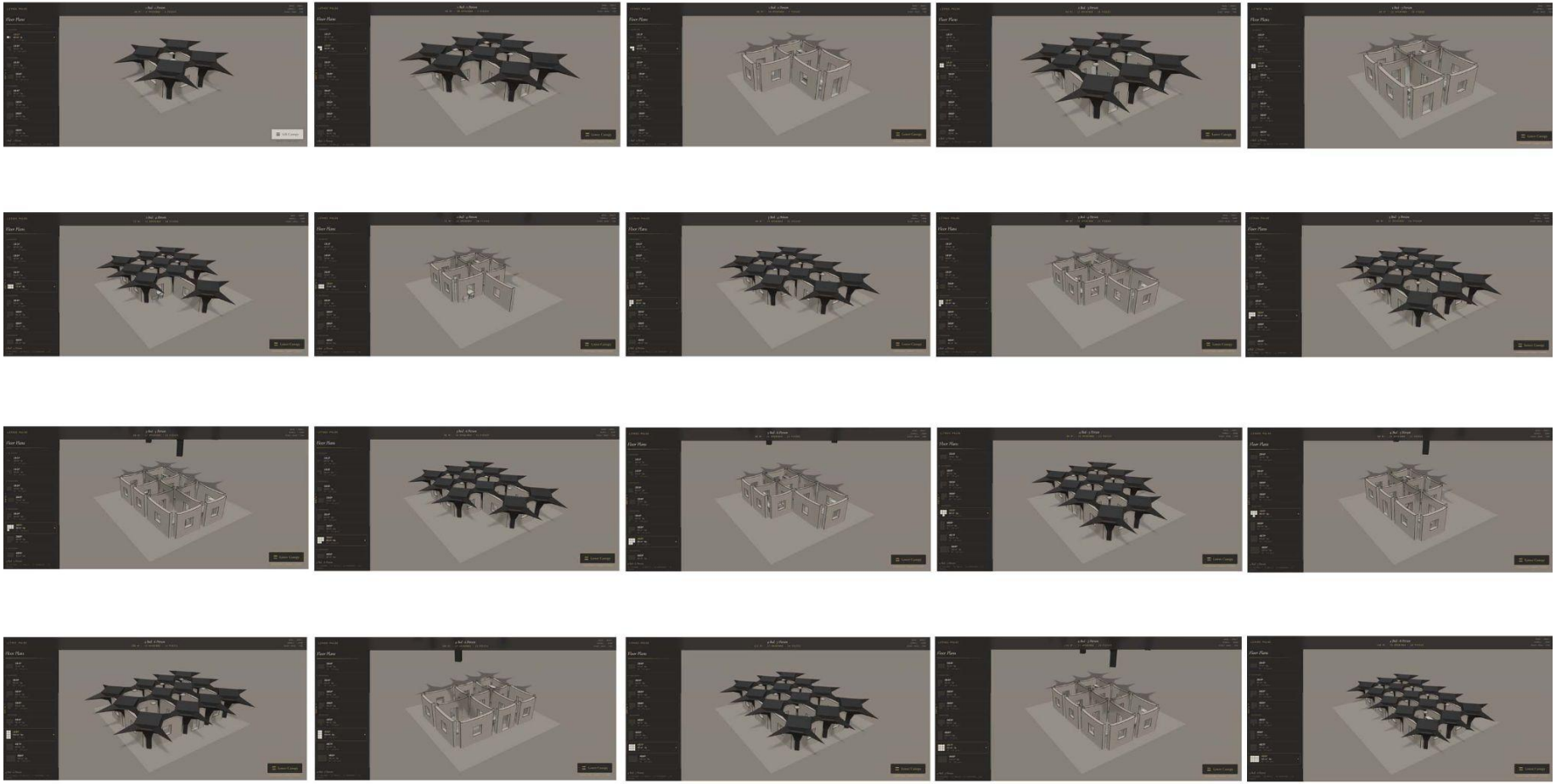


Figure 17. Units' design through computation
Source: Lithic Pulse RCX

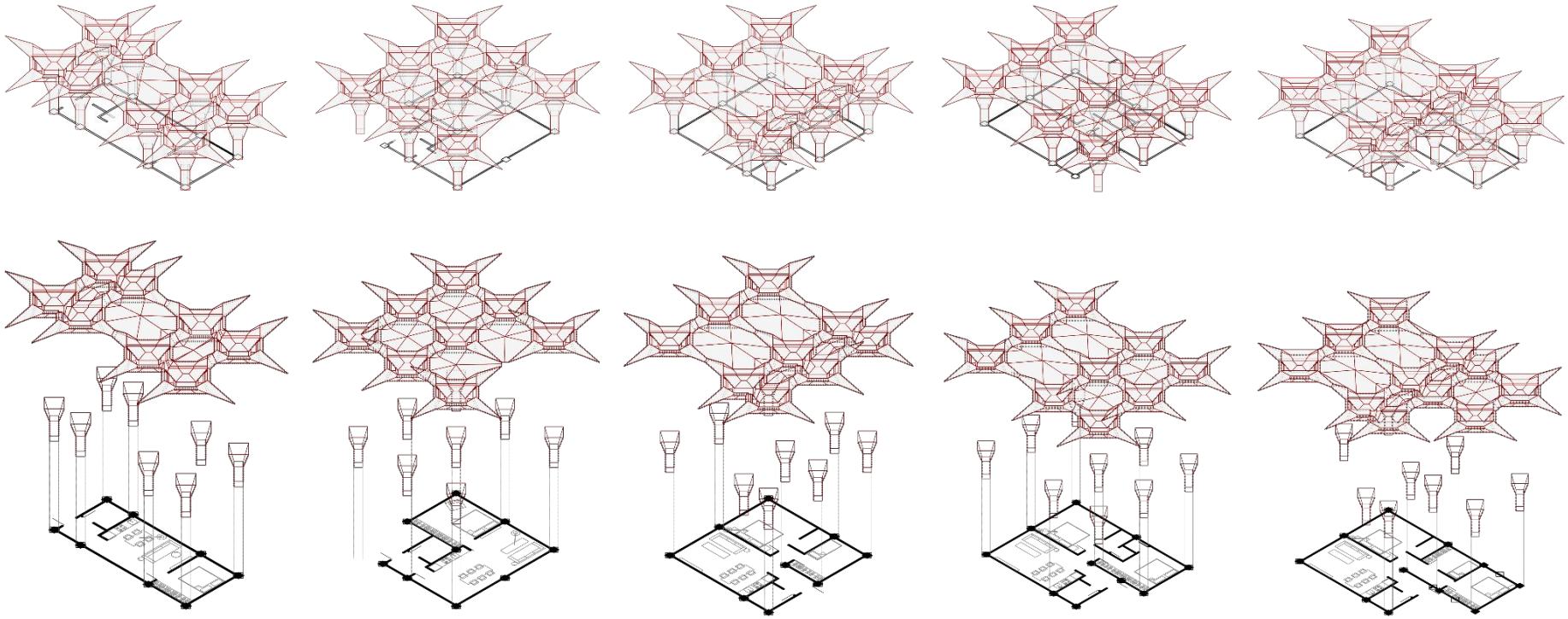
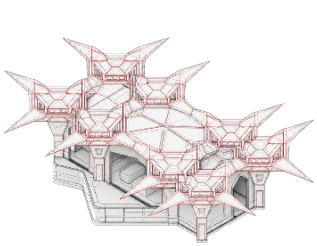
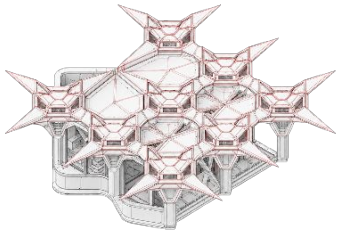


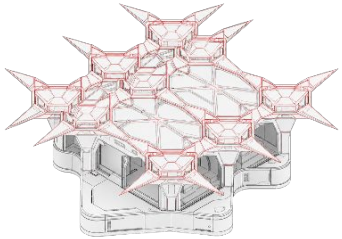
Figure 18. Generative design framework: user spatial planning, structural integration, and slab formation
Source: Lithic Pulse RCX



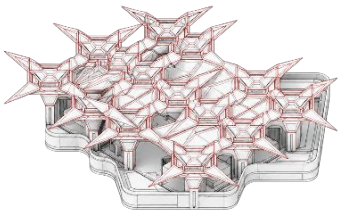
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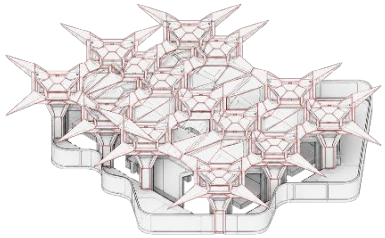
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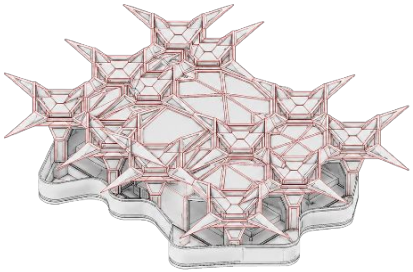
2b3p



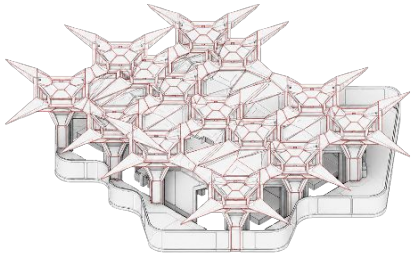
2b4p



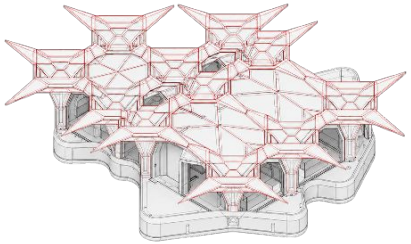
3b4p



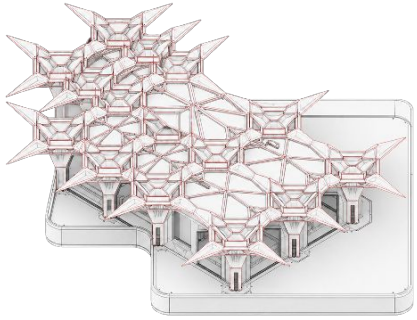
3b6p



4b5p



4b6p



4b7p

Figure 19. Units' design
Source: Lithic Pulse RCX

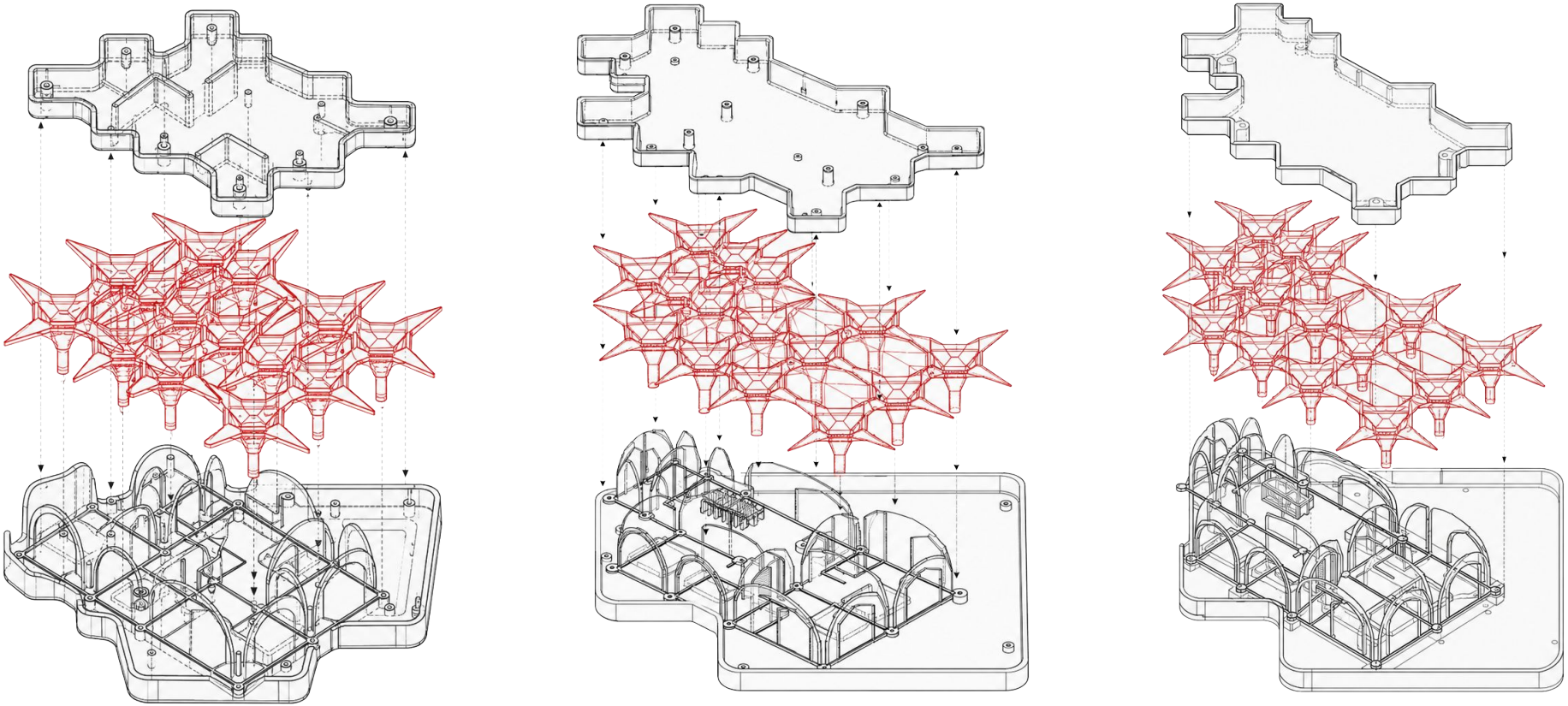


Figure 20. Exploded units' design for presenting kit-of-parts
Source: Lithic Pulse RCX





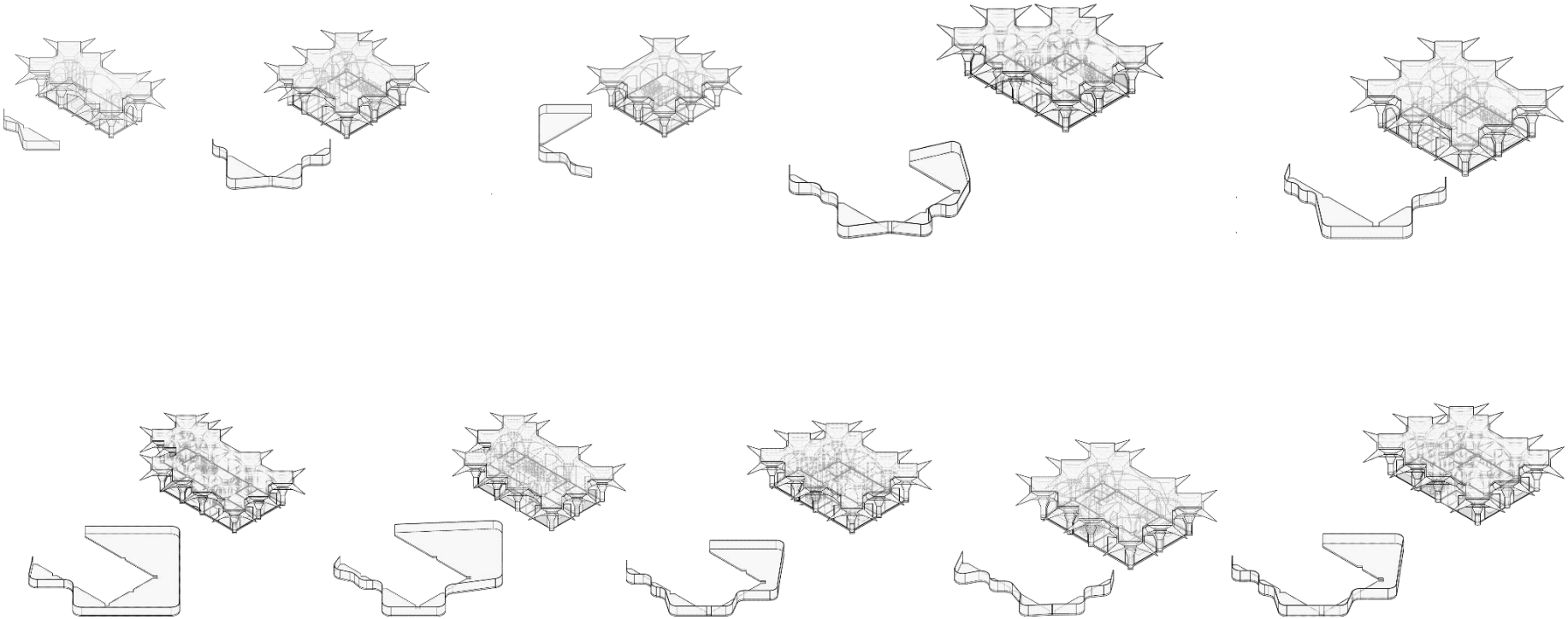


Figure 21. Kit-of-parts - balconies
Source: Lithic Pulse RCX

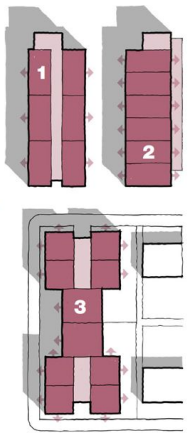
H.2.i Lighting aspect and privacy

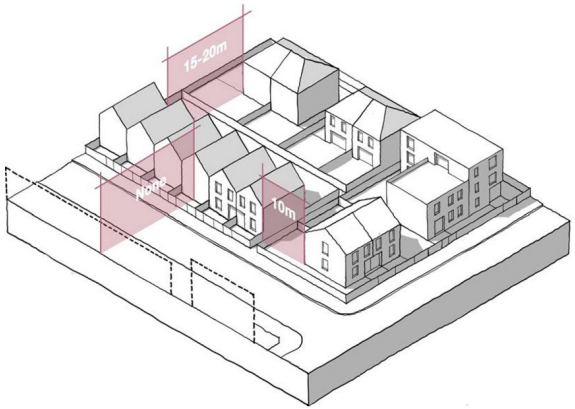
Dual aspect apartments

1. Single aspect flats should not face north.

2. Apartments accessed from a balcony have two aspects.

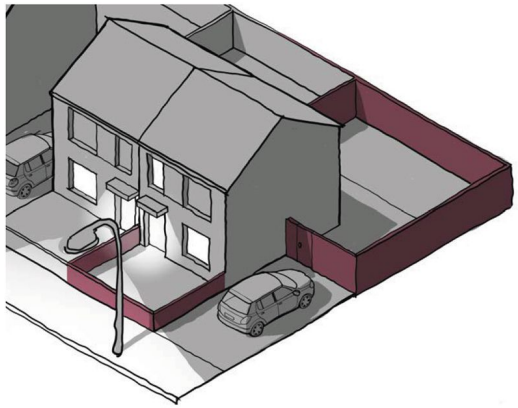
3. Walk-up apartments can also be designed so that every apartment has two aspects





3D line drawing of a residential block. It shows examples of the privacy distances set out in the preceding text. e.g. 10 meters between the side of one house and the rear of another, or 15-20 meters between the rear of two houses.

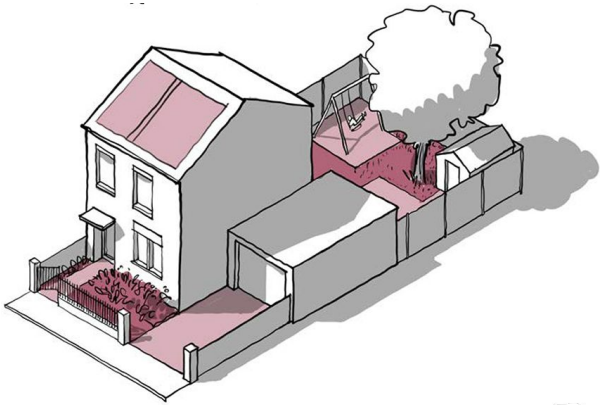
H.2.ii Security



3D line drawing of a semi detached house at night. It illustrates the principles set out in the preceding text on security.

Figure 22. Health and wellbeing design code for homes and building
Source: https://www.gov.uk

H.2.iii Gardens and balconies



3D line drawing of a house with a small front garden and larger back garden. The back garden contains a tree, play equipment and a small shed. It illustrates the preceding text on gardens and balconies.

Communal gardens

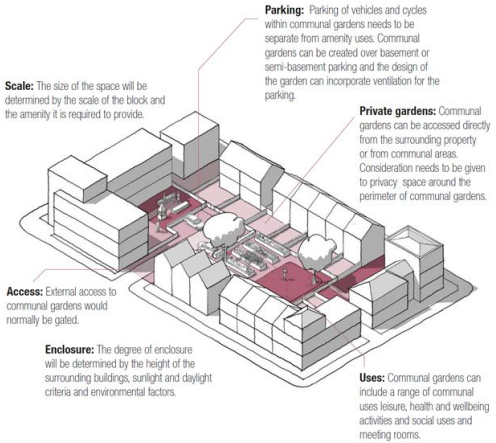
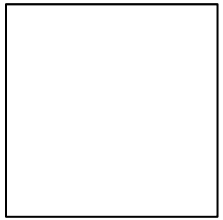
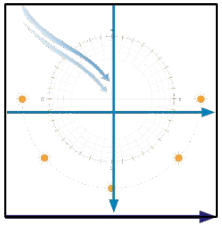


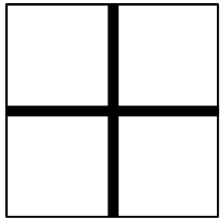
Figure 23. Health and wellbeing design code for homes and building
Source: https://www.gov.uk



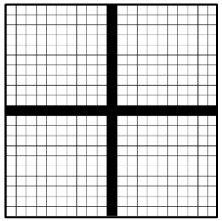
Select vacant site.



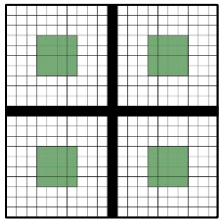
Analyze the site through the generated website, identifying sunlight exposure, prevailing wind directions, primary roads, and the main circulation routes across the site.



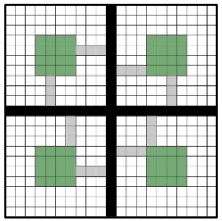
According to cluster user groups and main circulation direction divide site into 4 part.



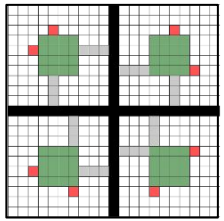
Divide each parts into 4x4x4 grid. Partition line should be main axis of site.



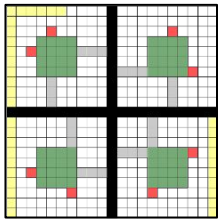
Define internal gardens for each user group. The internal garden area should constitute 10% of the allocated site area.



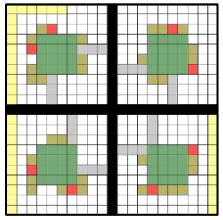
Indoor gardens should be connected to the main axis.



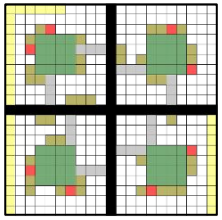
The vertical circulation core of the building is positioned parallel to the secondary axis.



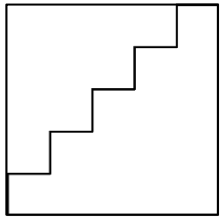
Public spaces should be aligned along the main road in order to mitigate noise and pollution impacts on the residential areas.



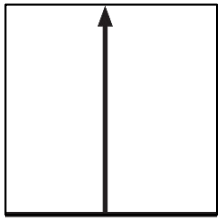
At least three private communal spaces should be designated for each user group, and these spaces should be oriented toward the internal gardens.



At least one shared communal space should be designated for each user group, and these spaces should be oriented along the main axis.



In response to solar orientation analyses, a system of stepped terraces is proposed for residential units across varying levels of the building mass. This architectural strategy maximizes daylight penetration and enhances natural ventilation within the interior spaces.



The building height should correspond to the arithmetic mean of the heights of the adjacent residential buildings.

Figure 24. Design codes for the spatial aggregation framework
Source: Lithic Pulse RCX



Figure 25. Brownfields and Vacant SItes throughout London
Source: Lithic Pulse RCX



Figure 26. Detailed Site Analysis for Site Selection and Optimised Planing
Source: Lithic Pulse RCX

Planning and space zoning according to site analysis

Segregation of units based on site and surroundings

Placement of units according to adjacency rules

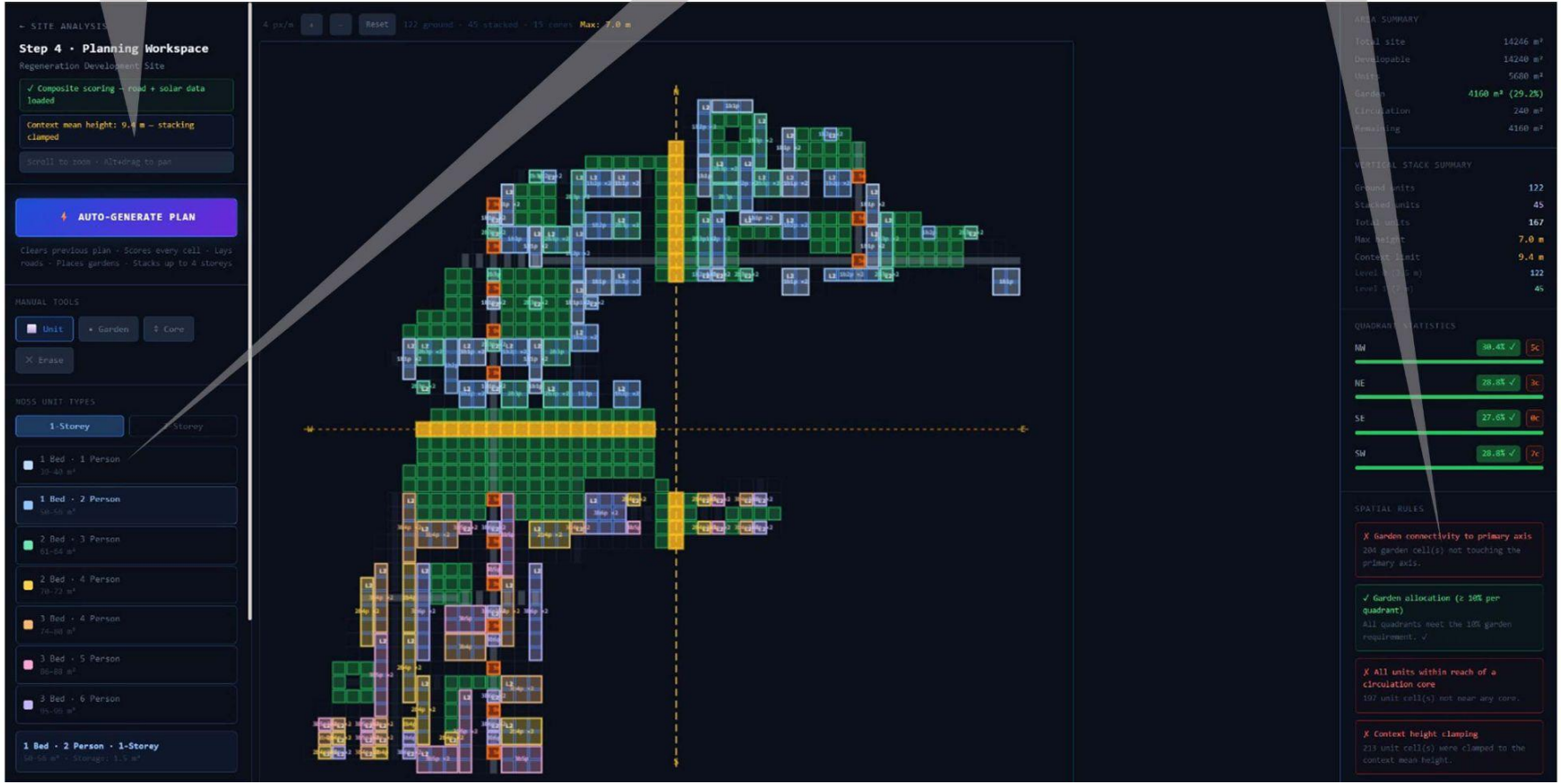


Figure 27. Spatially organized floor plans
Source: Lithic Pulse RCX

Planning and 3D massing according to site analysis

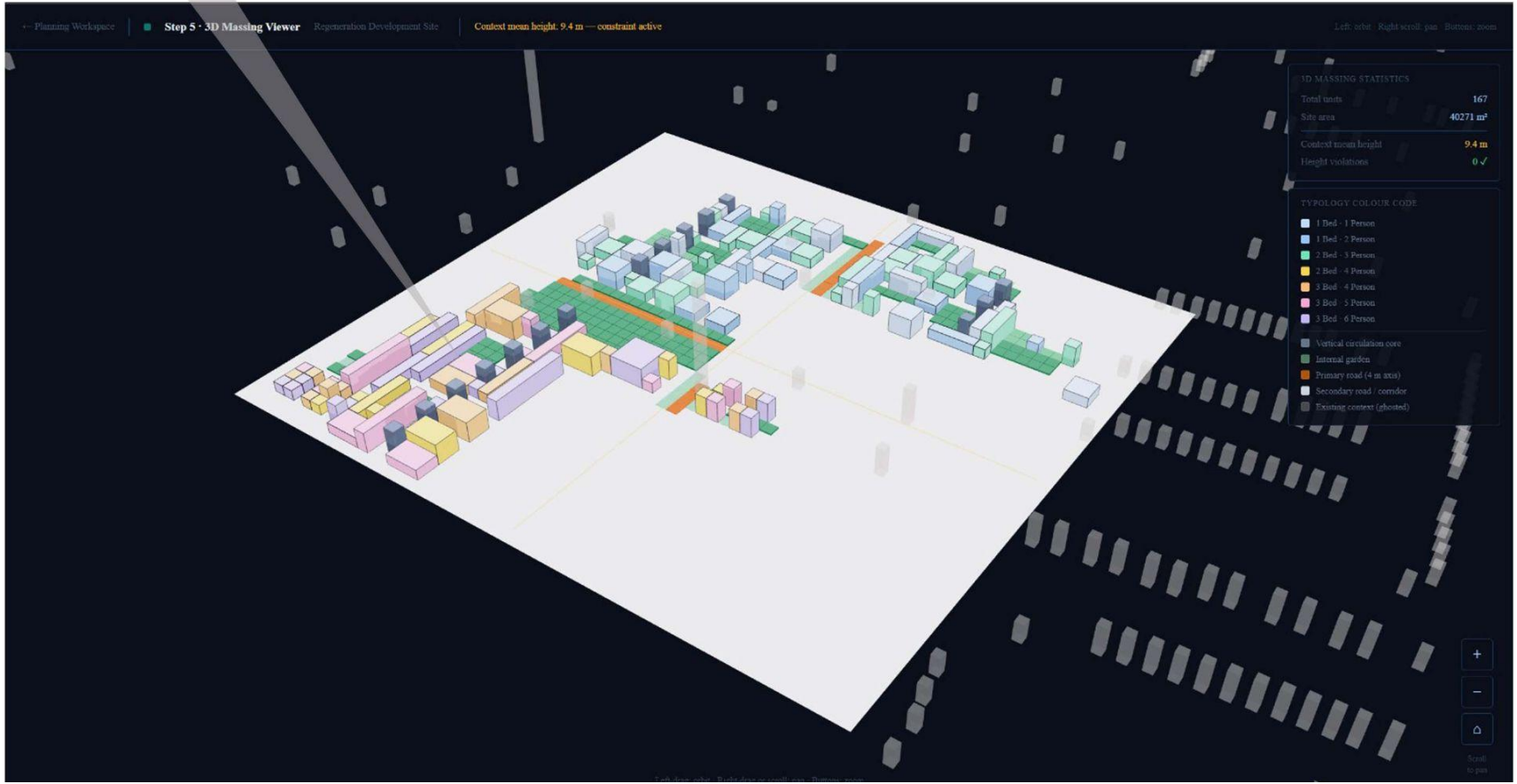


Figure 28. Spatially organized floor plans
Source: Lithic Pulse RCX

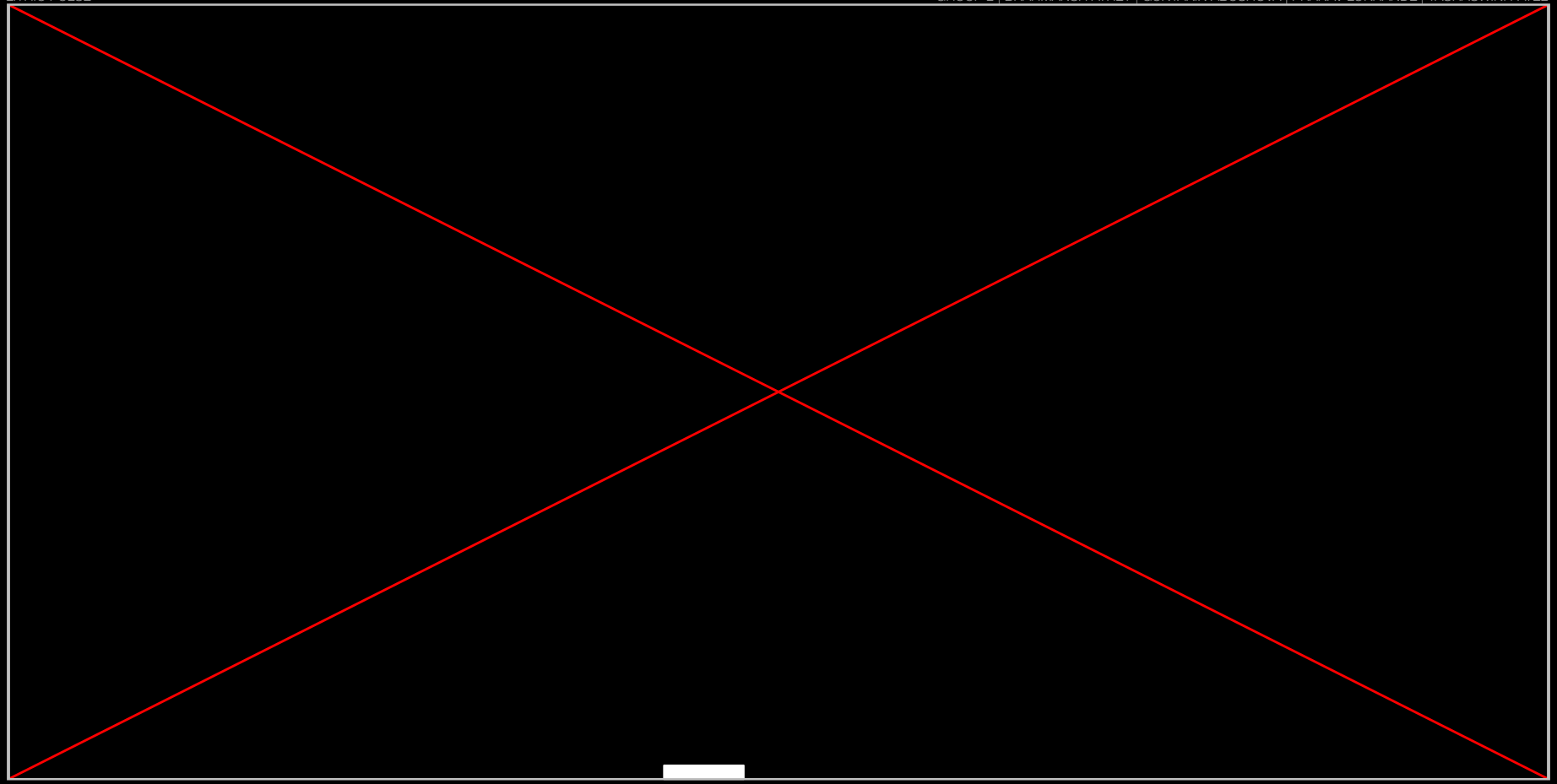


Figure 29. Spatially organized floor plans
Source: Lithic Pulse RCX

FABRICATION

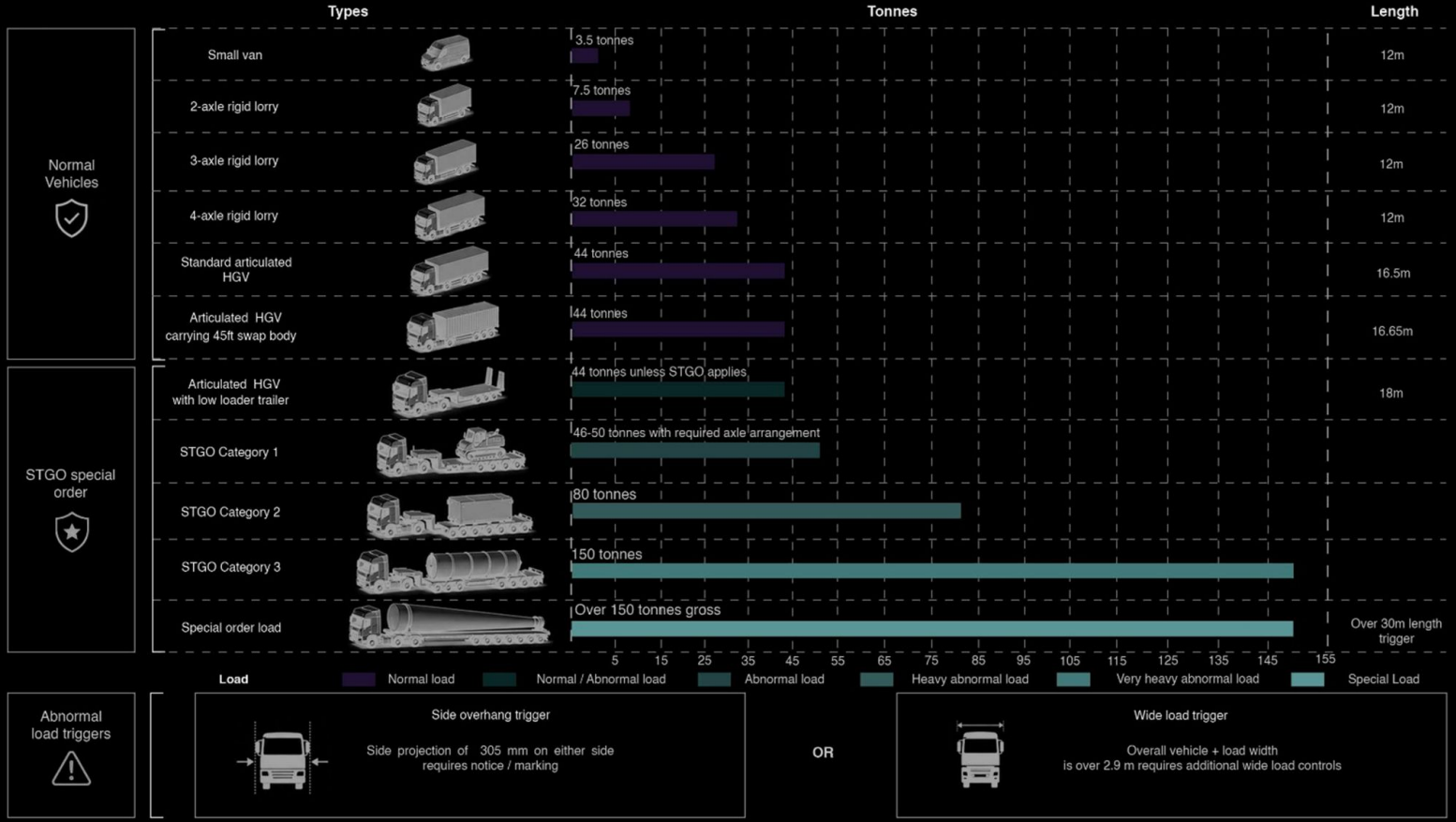


Figure 5. Truck rules by the UK Government

Source: Lithic Pulse RCX

GROUP 2 | BRAHMANSH ATREY | GUNTAKIN ABUSHOVA | PRANAV LOKHANDE | YASHASWINI PATEL

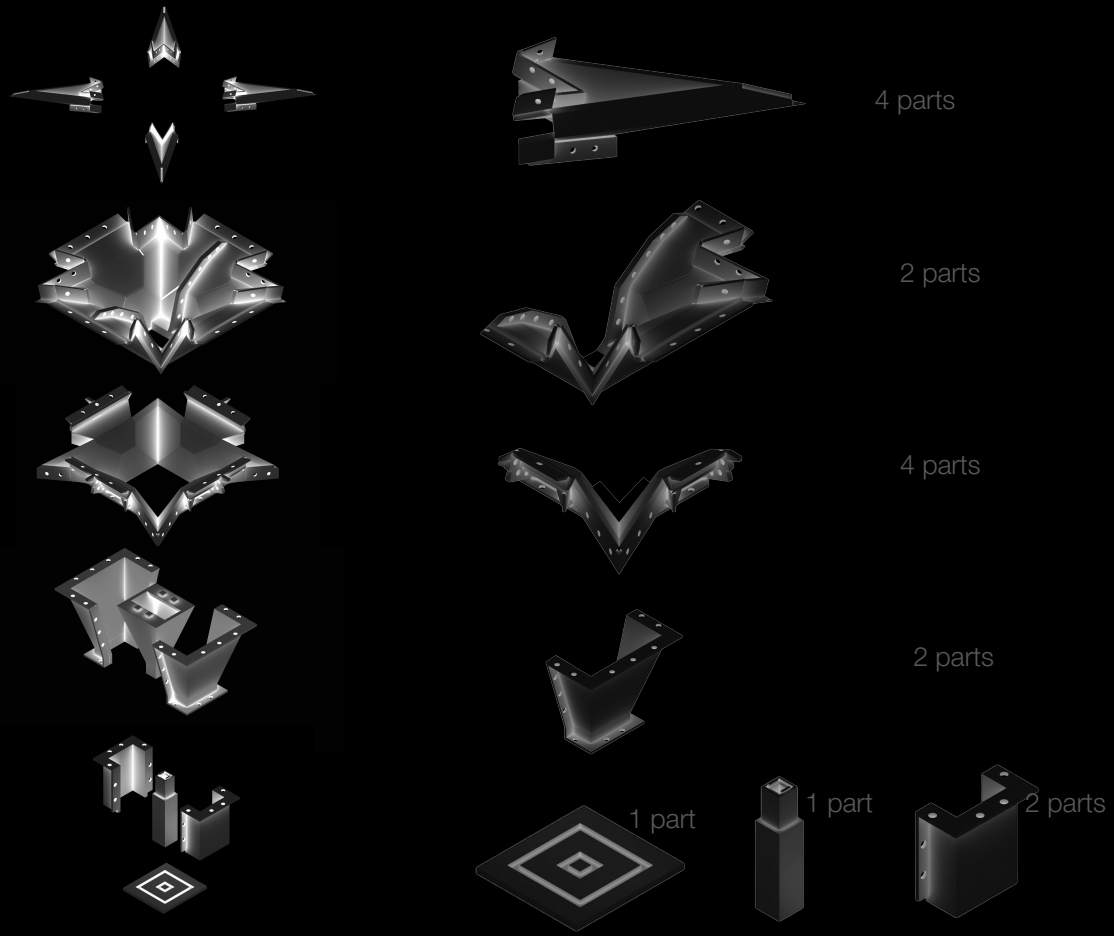
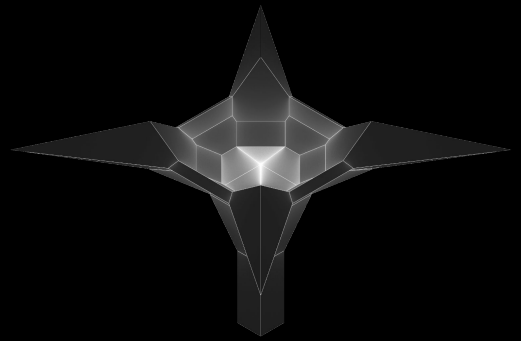
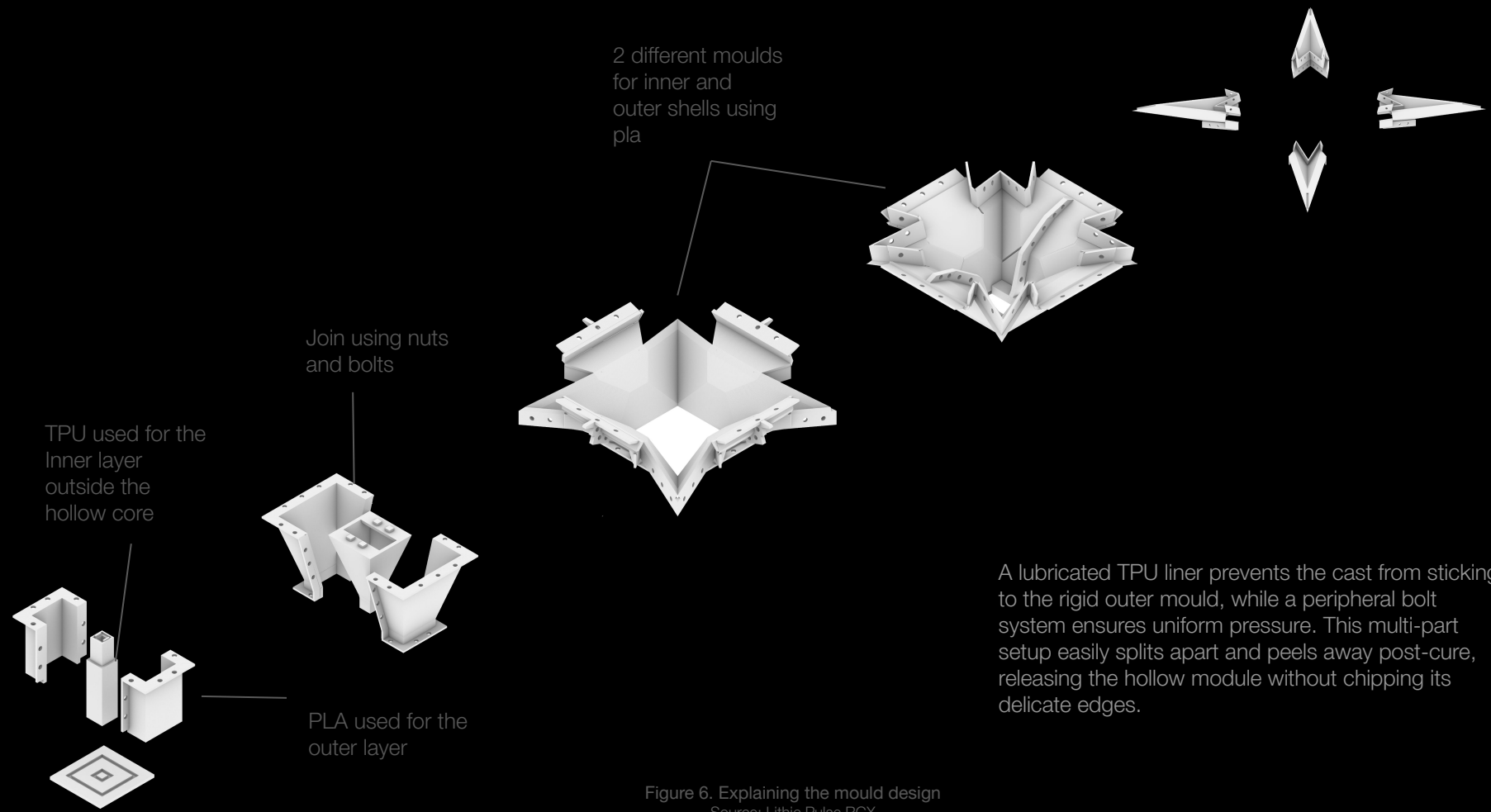


Figure 6. Mould parts
Source: Lithic Pulse RCX

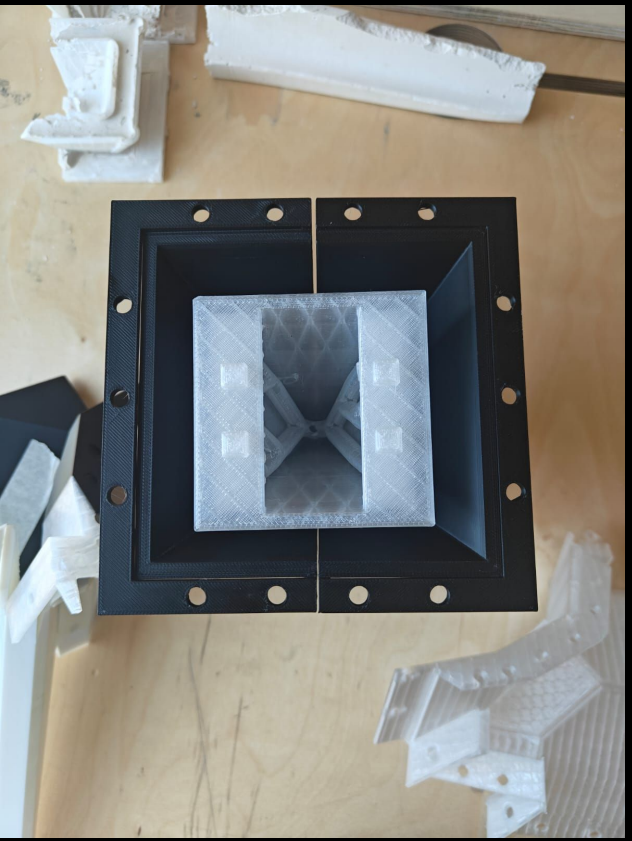


The formwork system is highly optimized, requiring only 8 unique master mould geometries to cast the entire structural sequence. By repeating these standard, interchangeable parts, production costs are drastically reduced while maintaining complete geometric consistency across the modular assembly.



A lubricated TPU liner prevents the cast from sticking to the rigid outer mould, while a peripheral bolt system ensures uniform pressure. This multi-part setup easily splits apart and peels away post-cure, releasing the hollow module without chipping its delicate edges.

Figure 6. Explaining the mould design
Source: Lithic Pulse RCX





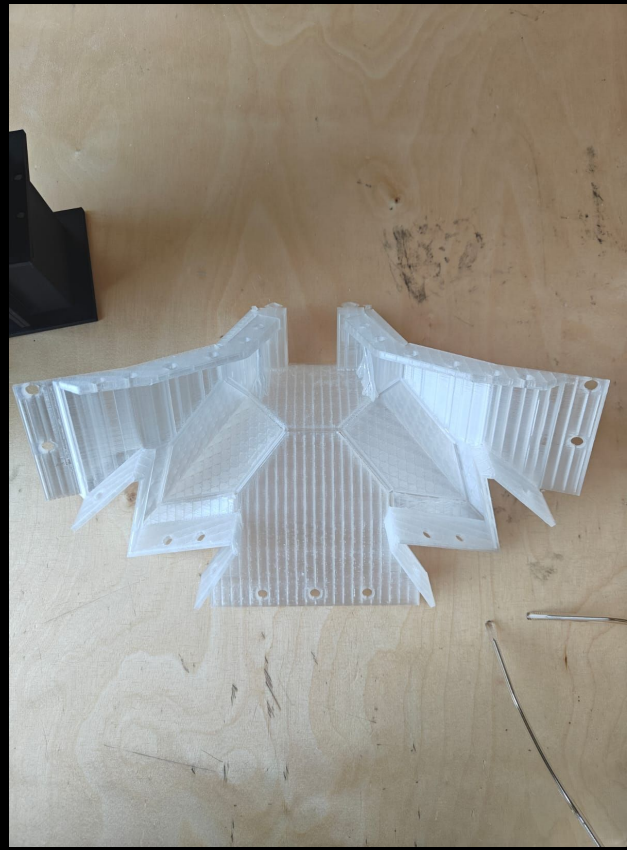
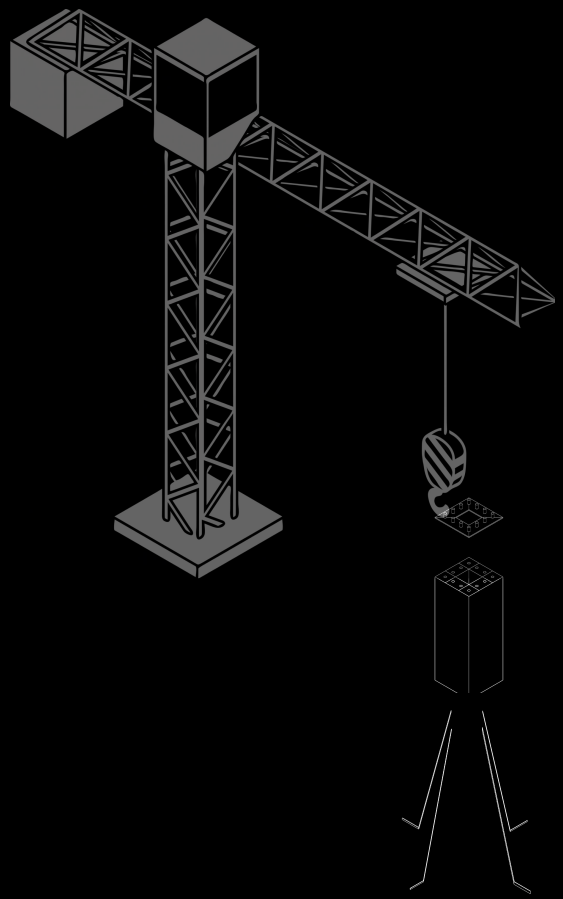
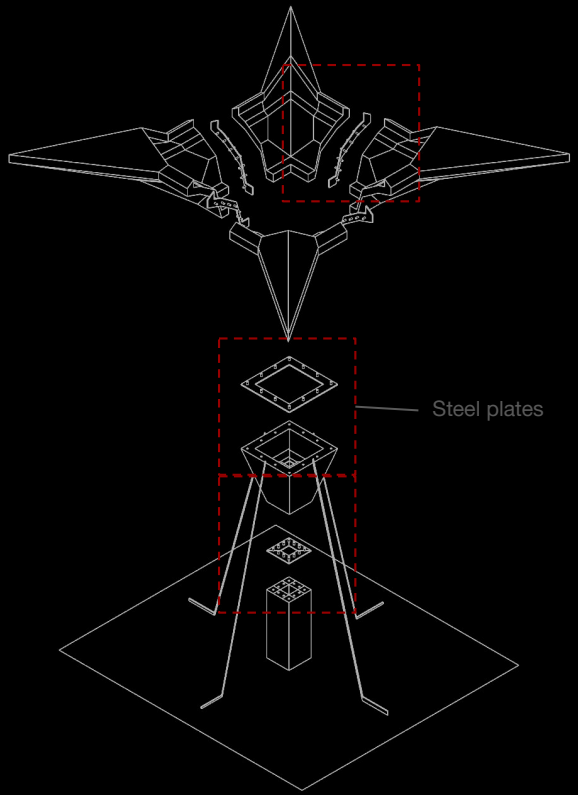


Figure 6. Physical moulds
Source: Lithic Pulse RCX





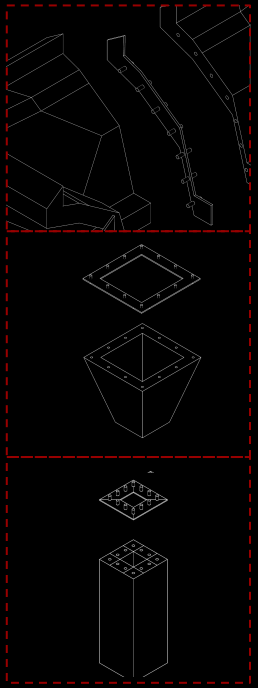


Figure 6. Assembly
Source: Lithic Pulse RCX

GROUP 2 | BRAHMANSH ATREY | GUNTAKIN ABUSHOVA | PRANAV LOKHANDE | YASHASWINI PATEL



GROUP 2 | BRAHMANSH ATREY | GUNTAKIN ABUSHOVA | PRANAV LOKHANDE | YASHASWINI PATEL









Thank you